Kernel Methods

Machine Learning – Foundations and Algorithms

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The “thing” about features

Good feature representations are typically very hard to find...

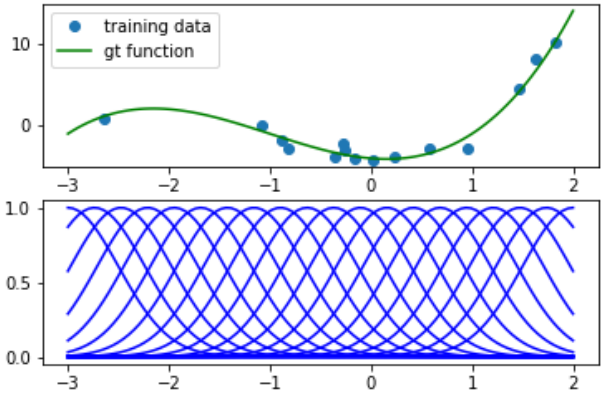
- **Polynomial features:**
  - Number of features increases with O(d^k), where d is dimensionality of data and k is degree of polynom
  - Can get very high feature value differences for large k

- **Radial basis function features:**
  - Uniformly distribute Gaussian basis functions over the input space

$$\boldsymbol{\phi}(\boldsymbol{x}) = [\phi_{\boldsymbol{\mu}_1}(\boldsymbol{x}), \ldots, \phi_{\boldsymbol{\mu}_M}(\boldsymbol{x})]^T, \quad \phi_{\boldsymbol{\mu}_i}(\boldsymbol{x}) = \exp\left(-\frac{||\boldsymbol{x} - \boldsymbol{\mu}_i||^2}{2\sigma^2}\right)$$

$\boldsymbol{\mu}_i \ldots$ i^{th} center location (fixed) $\quad \sigma^2 \ldots$ bandwith

  - Number of basis functions scales exponentially with d
  - Many basis functions located at irrelevant positions with no data



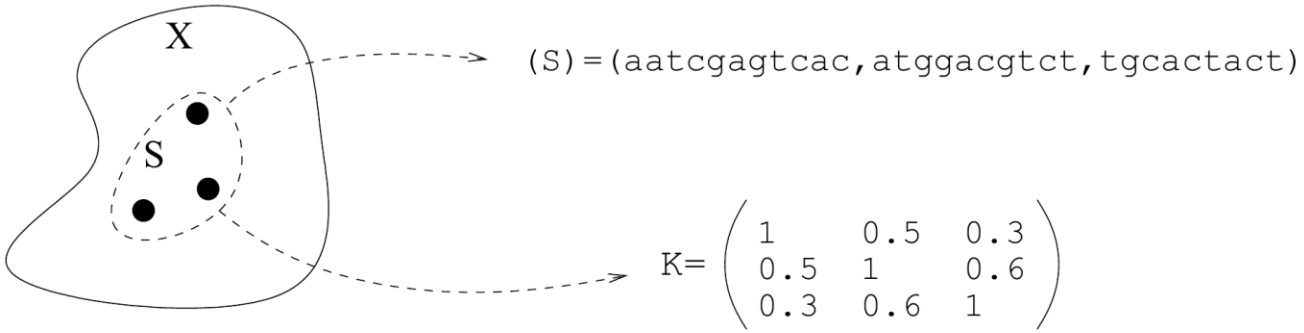
Kernels - Motivation

Can we find a more powerful representation for linear models?

- Yes! **Kernels!**
- They are more generic, scale better and adapt to the data set
- They still preserve simple linear computations

What is a kernel?

Representation by point-wise comparisons



- Define a "comparison function" $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$
- Represent a set of points $\mathcal{S} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ by **the n x n matrix** $[\boldsymbol{K}]_{ij} = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$

Kernel Matrix (aka Gram Matrix)

Properties:

- $\boldsymbol{K}$ is always an n × n matrix, whatever the nature of data: the same algorithm will work for any type of data (vectors, strings, ...).
- Total modularity between the choice of function k and the choice of the algorithm.
- Poor scalability with respect to the dataset size (n^2 to compute and store $\boldsymbol{K}$)...
- We will restrict ourselves to a particular class of pairwise comparison functions.

Positive definite kernels

A **positive definite kernel function** *k* is a function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ that is:

(i) Symmetric: $\forall \boldsymbol{x}, \boldsymbol{x}' : k(\boldsymbol{x}, \boldsymbol{x}') = k(\boldsymbol{x}', \boldsymbol{x})$

(ii) Similarity matrix is always positive semi-definite (psd)

$$\boldsymbol{a}^T \boldsymbol{K} \boldsymbol{a} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j k(\boldsymbol{x}_i, \boldsymbol{x}_j) \geq 0, \quad \forall \boldsymbol{a}, \forall S = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$$

- **Condition for psd matrices**: all eigenvalues >= 0

Kernel methods are algorithms that take such matrices as input.

Example: Linear kernel

The linear kernel is the simplest kernel for vectors

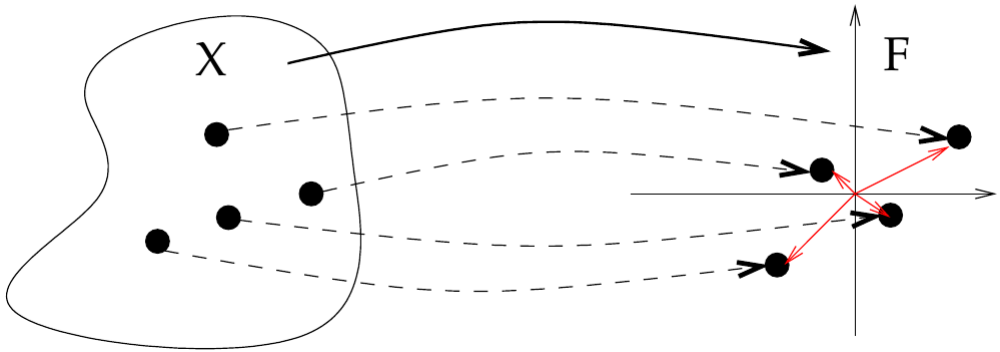
- Its defined by the scalar product:

$$k(\boldsymbol{x}, \boldsymbol{x}') = \langle \boldsymbol{x}, \boldsymbol{x}' \rangle, \quad \text{where } \langle \cdot, \cdot \rangle \text{ denotes the inner product}$$

- It is always positive definite:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \langle \boldsymbol{x}_i, \boldsymbol{x}_j \rangle = \Big\| \sum_i a_i \boldsymbol{x}_i \Big\|^2 \geq 0$$

Kernels in Feature Spaces



Let $\phi : \mathcal{X} \to \mathbb{R}^d$ be an arbitrary feature function, then $k(\boldsymbol{x}, \boldsymbol{x}') = \langle \boldsymbol{\phi}(\boldsymbol{x}), \boldsymbol{\phi}(\boldsymbol{x}') \rangle$ defines a positive definite kernel.

Proof: $$\sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \langle \boldsymbol{\phi}(\boldsymbol{x}_i), \boldsymbol{\phi}(\boldsymbol{x}_j) \rangle = \Big\| \sum_{i} a_i \boldsymbol{\phi}(\boldsymbol{x}_i) \Big\|^2 \geq 0$$

Kernels as inner products

Theorem (Aransjan 1950):

k is a positive semi-definite kernel on the set $\mathcal{X}$ if and only if there exists a feature space $\mathcal{H}$ and a feature mapping

$$\phi : \mathcal{X} \rightarrow \mathcal{H}$$

such that for any $\boldsymbol{x}, \boldsymbol{x}' \in \mathcal{X}$:

$$k(\boldsymbol{x}, \boldsymbol{x}') = \left\langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \right\rangle$$

- **Every psd kernel comes with an associated feature space!**

Kernel Regression

- What is a kernel?
- Examples for kernels
- The kernel trick
- Kernel Ridge Regression

Example: polynomial kernel

For $\boldsymbol{x} = [x_1, x_2]^T$, let $\boldsymbol{\phi}(\boldsymbol{x}) = [x_1^2, \sqrt{2}x_1x_2, x_2^2]$

The kernel is defined by:

$$\begin{aligned} k(\boldsymbol{x}, \boldsymbol{x}') &= x_1^2 x_1'^2 + 2x_1x_2x_1'x_2' + x_2^2 x_2'^2 \\ &= (x_1x_1' + x_2x_2')^2 \\ &= \langle \boldsymbol{x}, \boldsymbol{x}' \rangle^2 \end{aligned}$$

Kernel for polynomials of degree d:

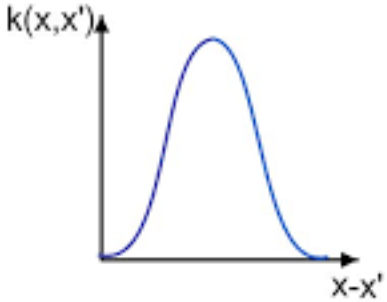
$$k(\boldsymbol{x}, \boldsymbol{x}') = \langle \boldsymbol{x}, \boldsymbol{x}' \rangle^d$$

Example: Gaussian Kernel

The Gaussian kernel is defined by:

$$k(\boldsymbol{x}, \boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{y}\|^2}{2\sigma^2}\right)$$

- where σ is the bandwidth parameter



Often also called:

- Radial basis function kernel (RBF)
- Squared exponential kernel

It is the most used kernel for kernel methods

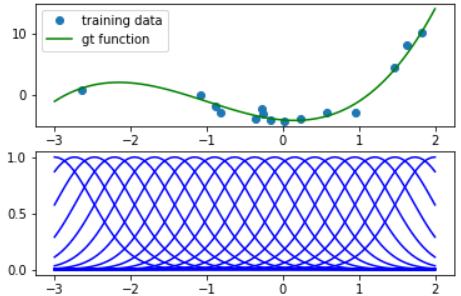
Is the Gaussian kernel a valid p.d. kernel?

Yes! It can be shown that the kernel is a valid product of two **infinite dimensional feature vectors**

- Consider the following feature function:

$$\phi_{\boldsymbol{\mu}}(\boldsymbol{x}) = 1/Z \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad \forall \boldsymbol{\mu} \in \mathbb{R}^d$$

- I.e. we have an infinite amount of features (for every possible center $\boldsymbol{\mu}$)
- Z is a normalization constant (which we will ignore)



The Gaussian kernel would put a RBF center at every possible location

The Gaussian kernel can be computed as (not shown)**:**

$$k(\boldsymbol{x},\boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{y}\|^2}{2\sigma^2}\right) = \langle \phi_{\boldsymbol{\mu}}(\boldsymbol{x}), \phi_{\boldsymbol{\mu}}(\boldsymbol{y}) \rangle = \int \phi_{\boldsymbol{\mu}}(\boldsymbol{x})\phi_{\boldsymbol{\mu}}(\boldsymbol{y}) d\boldsymbol{\mu}$$

- Inner product becomes an integral due to infinite amount of dimensions
- We never have to represent these vectors if we can directly compute their inner products

Kernel Regression

- What is a kernel?
- Examples for kernels
- The kernel trick
- Kernel Ridge Regression

Kernel Trick

So why do we do this?

- Kernels can be used for most feature based algorithms that can be rewritten such that they contain inner products of feature vectors
  - This is true for almost all feature based algorithms (Linear regression, Support Vector Machines, …)
  - This is called the Kernel Trick
- Kernels can be used to map the data x in an infinite dimensional feature space (i.e., a function space)
  - The feature vector never has to be represented explicitly
  - As long as we can evaluate the inner product of two feature vectors
- Hence, we obtain a more powerful representation than standard linear feature models

A few kernel identities

Let $\boldsymbol{\Phi}_X = \begin{bmatrix} \boldsymbol{\phi}(\boldsymbol{x}_1)^T \\ \vdots \\ \boldsymbol{\phi}(\boldsymbol{x}_N)^T \end{bmatrix} \in \mathbb{R}^{N \times d}$ then the following identities hold:

- **Kernel matrix:** $\boldsymbol{K} = \boldsymbol{\Phi}_X \boldsymbol{\Phi}_X^T$
  - Check: $[\boldsymbol{K}]_{ij} = \boldsymbol{\phi}(\boldsymbol{x}_i)^T \boldsymbol{\phi}(\boldsymbol{x}_j) = k(\boldsymbol{x}_i, \boldsymbol{x}_j)$
- **Kernel vector:** $\boldsymbol{k}(\boldsymbol{x}^*) = \begin{bmatrix} k(\boldsymbol{x}_1, \boldsymbol{x}^*) \\ \vdots \\ k(\boldsymbol{x}_N, \boldsymbol{x}^*) \end{bmatrix} = \begin{bmatrix} \boldsymbol{\phi}(\boldsymbol{x}_1)^T \boldsymbol{\phi}(\boldsymbol{x}^*) \\ \vdots \\ \phi(\boldsymbol{x}_N)^T \boldsymbol{\phi}(\boldsymbol{x}^*) \end{bmatrix} = \boldsymbol{\Phi}_X \boldsymbol{\phi}(\boldsymbol{x}^*)$

Kernel trick

Recap: Ridge Regression

- Squared error function + L2 regularization
- Linear feature space
- Not directly applicable in infinite dimensional feature spaces

Objective:

$$L_{\text{ridge}} = \underbrace{(\boldsymbol{y} - \boldsymbol{\Phi w})^T (\boldsymbol{y} - \boldsymbol{\Phi w})}_{\text{sum of squared errors}} + \lambda \underbrace{\boldsymbol{w}^T \boldsymbol{w}}_{L_2 \text{ regularization}}$$

Solution:

$$\boldsymbol{w}^*_{\text{ridge}} = \underbrace{(\boldsymbol{\Phi}^T \boldsymbol{\Phi} + \lambda \boldsymbol{I})^{-1}}_{d \times d \text{ matrix inversion}} \boldsymbol{\Phi}^T \boldsymbol{y}$$

Matrix inversion infeasible in infinite dimensions

Kernel trick

We can apply the "kernel trick":

- Rewrite solution as inner products of the feature space!
- We can do this by using the following matrix identity

$$(\lambda \boldsymbol{I} + \boldsymbol{AB})^{-1}\boldsymbol{A} = \boldsymbol{A}(\lambda \boldsymbol{I} + \boldsymbol{BA})^{-1}$$

  - "Searle set of identities", The Matrix Cookbook

$$\boldsymbol{w}^* = \underbrace{(\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \lambda \boldsymbol{I})^{-1}}_{d \times d \text{ matrix inversion}} \boldsymbol{\Phi}^T \boldsymbol{y} = \boldsymbol{\Phi}^T \underbrace{(\boldsymbol{\Phi}\boldsymbol{\Phi}^T + \lambda \boldsymbol{I})^{-1}}_{N \times N \text{ matrix inversion}} \boldsymbol{y}$$

  - With $\boldsymbol{A} = \boldsymbol{\Phi}^T$ and $\boldsymbol{B} = \boldsymbol{\Phi}$

The „tale of the two Pseudo Inverse“

- **Two side inverse:** $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}, \quad \boldsymbol{A}\boldsymbol{A}^{-1} = \boldsymbol{I}$
  - Standard matrix inversion… does often not exist
- **Left-Pseudo Inverse (see linear regression):**

$$\boldsymbol{A}^{\dagger}_{\text{left}} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T, \quad \boldsymbol{A}^{\dagger}_{\text{left}}\boldsymbol{A} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}$$

- **Right-Pseudo Inverse:**

$$\boldsymbol{A}^{\dagger}_{\text{right}} = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1}, \quad \boldsymbol{A}\boldsymbol{A}^{\dagger}_{\text{right}} = \boldsymbol{A}\boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T)^{-1} = \boldsymbol{I}$$

Difference: Size of the matrix that needs to be inverted

- For regularized version, both are mathematically (but not numerically) equivalent

$$(\boldsymbol{A}^T\boldsymbol{A} + \lambda\boldsymbol{I})^{-1}\boldsymbol{A}^T = \boldsymbol{A}^T(\boldsymbol{A}\boldsymbol{A}^T + \lambda\boldsymbol{I})^{-1}$$

Kernel Regression

- What is a kernel?
- Examples for kernels
- The kernel trick
- Kernel Ridge Regression

Kernel ridge regression

The "kernelized" solution is given by:

$$\boldsymbol{w}^* = \boldsymbol{\Phi}^T \underbrace{(\boldsymbol{\Phi}\boldsymbol{\Phi}^T + \lambda \boldsymbol{I})^{-1}}_{N \times N \text{ matrix inversion}} \boldsymbol{y} = \boldsymbol{\Phi}^T \underbrace{(\boldsymbol{K} + \lambda \boldsymbol{I})^{-1}\boldsymbol{y}}_{\boldsymbol{\alpha}} = \boldsymbol{\Phi}^T \boldsymbol{\alpha}$$

- Instead of inverting a d x d matrix, we can now invert an N x N matrix
- Is beneficial for d >> N (e.g., d is infinite)
- Still, $\boldsymbol{w}^* \in \mathbb{R}^d$ is potentially infinite dimensional and can not be represented

Yet, we can evaluate the function f that is specified by $\boldsymbol{w}^*$:

$$f(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{w}^* = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\Phi}^T \boldsymbol{\alpha} = \boldsymbol{k}(\boldsymbol{x})^T \boldsymbol{\alpha} = \sum_i \alpha_i k(\boldsymbol{x}_i, \boldsymbol{x})$$

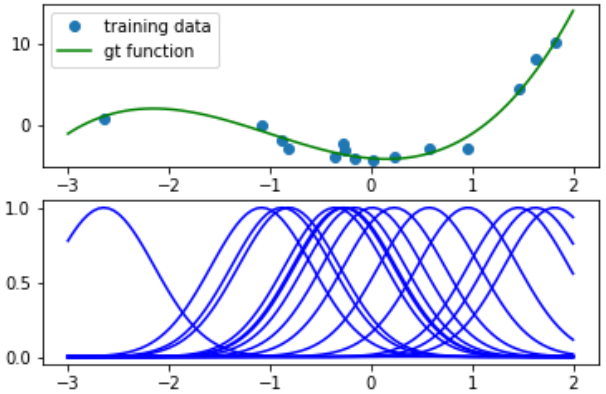
- The function is evaluated by a weighted sum of kernel activations evaluated at the training data

Examples and comparison to RBF regression

For a Gaussian kernel, the prediction corresponds to

$$f(\boldsymbol{x}) = \sum_i \alpha_i k(\boldsymbol{x}_i, \boldsymbol{x}) = \sum_i \alpha_i \exp\left(-\frac{||\boldsymbol{x} - \boldsymbol{x}_i||^2}{2\sigma^2}\right)$$

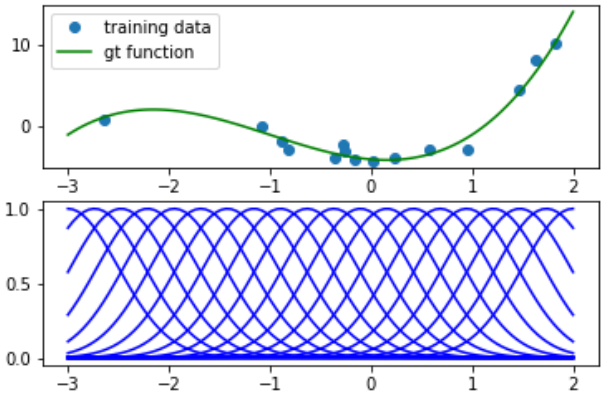
- The kernel allows setting the centres adaptively to the available data!
- One centre per data-point



Comparison: Linear regression with radial basis function (RBF) features

$$f(\boldsymbol{x}) = \sum_i w_i \phi_i(\boldsymbol{x}) = \sum_i w_i \exp\left(-\frac{||\boldsymbol{x} - \boldsymbol{\mu}_i||^2}{2\sigma^2}\right)$$

$\boldsymbol{\mu}_i \ldots$ i^{th} center location (fixed)



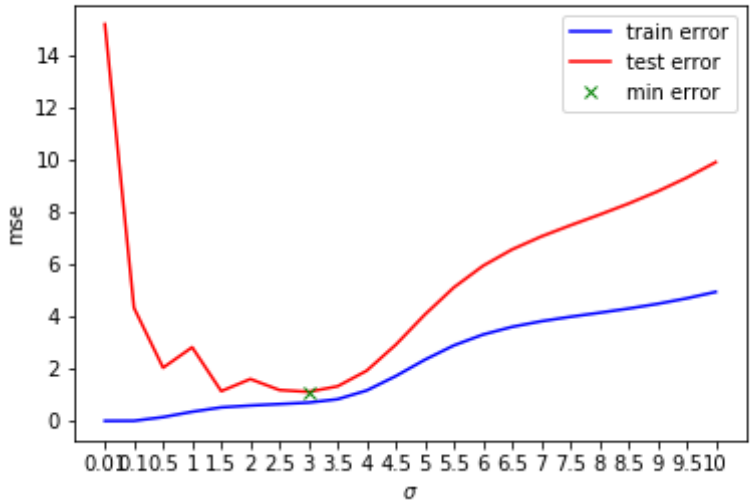
Selecting the hyper-parameters

- The parameters of the kernel, e.g., sigma in

$$k(\boldsymbol{x}, \boldsymbol{y}) = \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{y}\|^2}{2\sigma^2}\right)$$

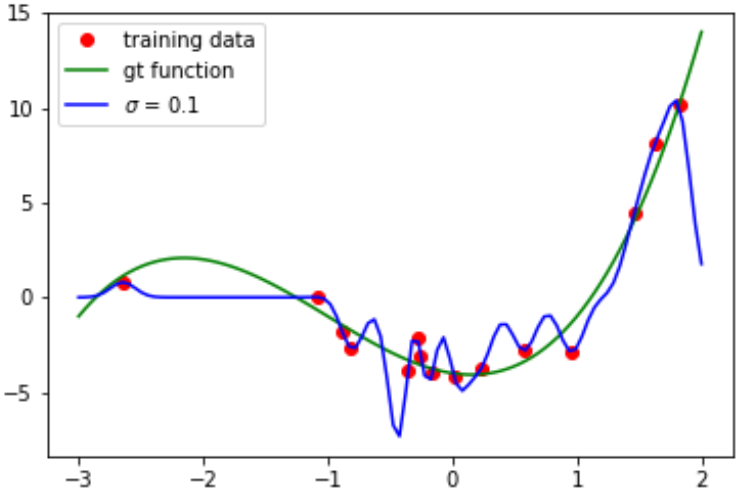
  are called hyper-parameters.

- Choosing them is a model-selection problem (we need a test set).

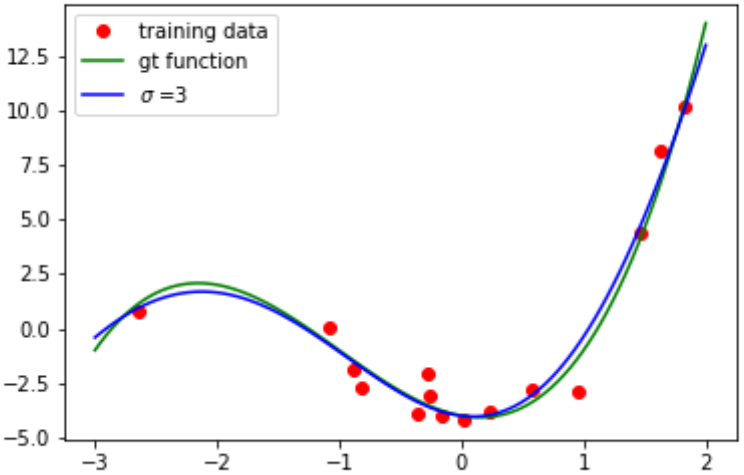


Different bandwidth factors

Overfitting



Good fit



Comparison

- **Ridge Regression**

$$f(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{w}^* = \boldsymbol{\phi}(\boldsymbol{x})^T \underbrace{(\boldsymbol{\Phi}^T \boldsymbol{\Phi} + \lambda \boldsymbol{I})^{-1}}_{d \times d \text{ matrix inversion}} \boldsymbol{\Phi}^T \boldsymbol{y} = \sum_{i=1}^{d} w_i \phi_i(\boldsymbol{x})$$

  - Use if d < N

- **Kernel Ridge Regression**

$$f(\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{w}^* = \overbrace{\boldsymbol{\phi}(\boldsymbol{x})^T \boldsymbol{\Phi}}^{\boldsymbol{k}(\boldsymbol{x})} \underbrace{(\overbrace{\boldsymbol{\Phi} \boldsymbol{\Phi}^T}^{\boldsymbol{K}} + \lambda \boldsymbol{I})^{-1}}_{N \times N \text{ matrix inversion}} \boldsymbol{y} = \sum_{i=1}^{N} \alpha_i k(\boldsymbol{x}, \boldsymbol{x}_i)$$

  - Use if N < d

Summary: Kernel ridge regression

The solution for kernel ridge regression is given by

$$f^*(\boldsymbol{x}) = \boldsymbol{k}(\boldsymbol{x})^T(\boldsymbol{K} + \lambda \boldsymbol{I})^{-1}\boldsymbol{y}$$

- No evaluations of the feature vectors needed
- Only pair-wise scalar products (evaluated by the kernel)
- Need to invert a N x N matrix (can be costly

Note:

- We have to store all samples in kernel-based methods (they also belong to the instance-based or non-parametric methods)
  - Computationally expensive (matrix inverse is $O(n^{2.376})$) !
- Hyper-parameters of the method are given by the kernel-parameters
  - Can be optimized on validation-set
- Very flexible function representation, only few hyper-parameters

SVMs with Kernels

Support Vector Machines (continued…)

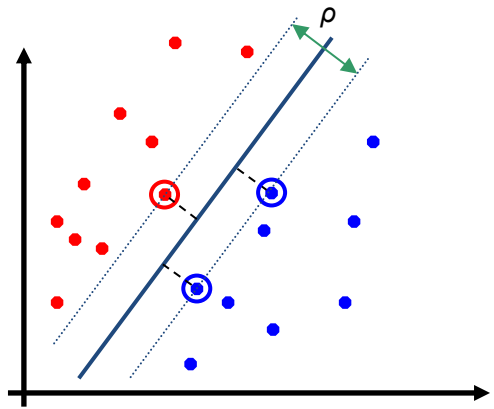
SVMs with features:

- Maximum margin principle
- Slack variables allow for margin violation

$$\begin{aligned} \text{argmin}_{\mathbf{w},\boldsymbol{\xi}} \quad & ||\mathbf{w}||^2 + C\sum_i^N \xi_i, \\ \text{s.t.} \quad & y_i(\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_i) + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \end{aligned}$$

- Simpler formulation without slack variables

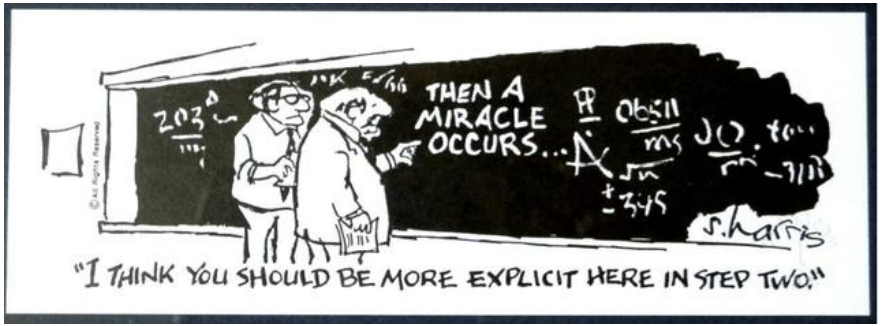
$$\begin{aligned} \text{argmin}_{\mathbf{w}} \quad & ||\mathbf{w}||^2, \\ \text{s.t.} \quad & y_i(\mathbf{w}^T\boldsymbol{\phi}(\mathbf{x}_i) + b) \geq 1 \end{aligned}$$



In order to apply the kernel trick, we need to apply Constrained Optimization!

- … however, we do not have time for going into that
- See RL lecture

Constraint Optimization in SVMs



In constraint optimization, we have to optimize the dual function:

$$\boldsymbol{\lambda}^* = \text{argmax}_{\boldsymbol{\lambda}} g(\boldsymbol{\lambda}) = \text{argmax}_{\boldsymbol{\lambda}} \sum_i \lambda_i - \frac{1}{2} \sum_i \sum_j \lambda_i \lambda_j y_i y_j \boldsymbol{\phi}(\boldsymbol{x}_i)^T \boldsymbol{\phi}(\boldsymbol{x}_j), \text{ s.t. } \lambda_i > 0$$

- Where λ_i are Lagrangian multipliers
- The weight vector can then be obtained by

$$\boldsymbol{w}^* = \sum_i \lambda_i y_i \boldsymbol{\phi}(\boldsymbol{x}_i)$$

We can directly use the kernel trick here

$$g(\boldsymbol{\lambda}) = \sum_i \lambda_i - \frac{1}{2} \sum_i \sum_j \lambda_i \lambda_j y_i y_j \boldsymbol{k}(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

- Scalar products of the feature vectors can be written as kernels, i.e. $k(\boldsymbol{x}_i, \boldsymbol{x}_j) = \boldsymbol{\phi}(\boldsymbol{x}_i)^T \boldsymbol{\phi}(\boldsymbol{x}_j)$

Relaxed constraints with slack

Primal optimization problem:

$$\begin{aligned} \text{argmin}_{\mathbf{w},\boldsymbol{\xi}} \quad & ||\mathbf{w}||^2 + C\sum_{i}^{N}\xi_i, \\ \text{s.t.} \quad & y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \end{aligned}$$

Dual optimization problem:

$$\max_{\boldsymbol{\lambda}} \sum_i \lambda_i - \frac{1}{2}\sum_i\sum_j \lambda_i\lambda_j y_i y_j \boldsymbol{k}(\boldsymbol{x}_i, \boldsymbol{x}_j)$$

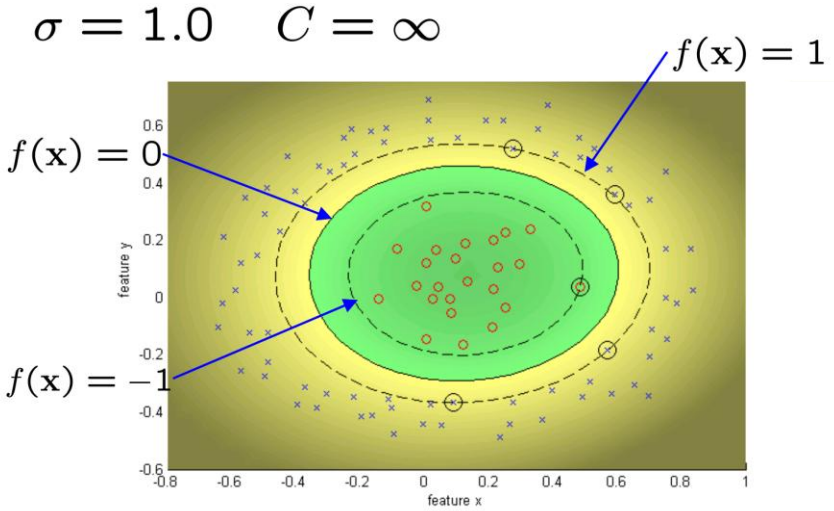
$$\text{s.t. } C \geq \lambda_i \geq 0, \forall i \in [1 \ldots N], \quad \sum_i \lambda_i y_i = 0$$

This is a quadratic program

- Quadratic objective, linear constraints
- Efficient solution solvers exists

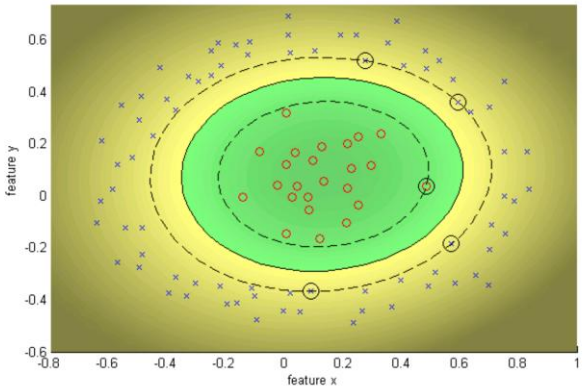
Example: SVM with RBF kernel

Data is non-linearly separable in original space

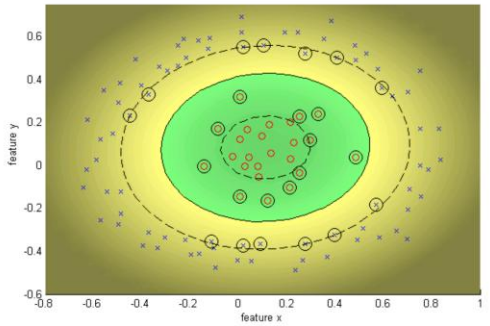


Example: Different Cs

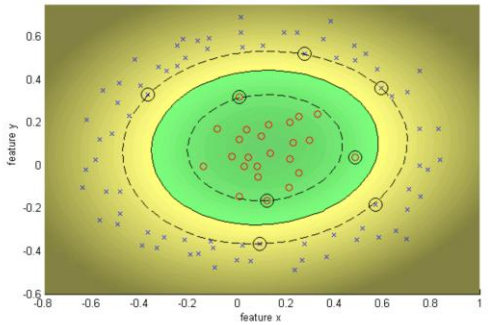
$\sigma = 1.0 \quad C = \infty$



$\sigma = 1.0 \quad C = 10$

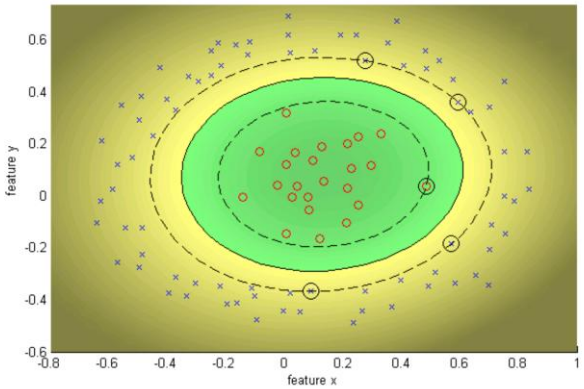


$\sigma = 1.0 \quad C = 100$

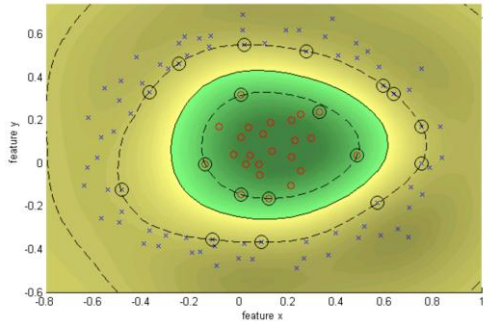


Example: Different sigma

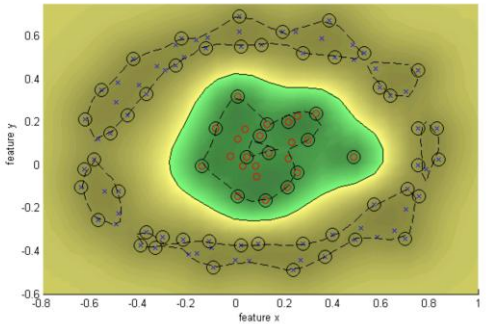
$\sigma = 1.0 \quad C = \infty$



$\sigma = 0.25 \quad C = \infty$



$\sigma = 0.1 \quad C = \infty$



Overfitting

Huge feature space with kernels: should we worry about overfitting?

- SVM objective seeks a solution with large margin
- Theory says that large margin leads to good generalization
- But everything overfits sometimes!!!

Can control overfitting by:

- Setting C (low C -> smaller Complexity)
- Choosing a better Kernel
- Varying parameters of the Kernel (width of Gaussian, etc.)

} Model Selection Problems

Handwritten Digit Clasification

US postal service database

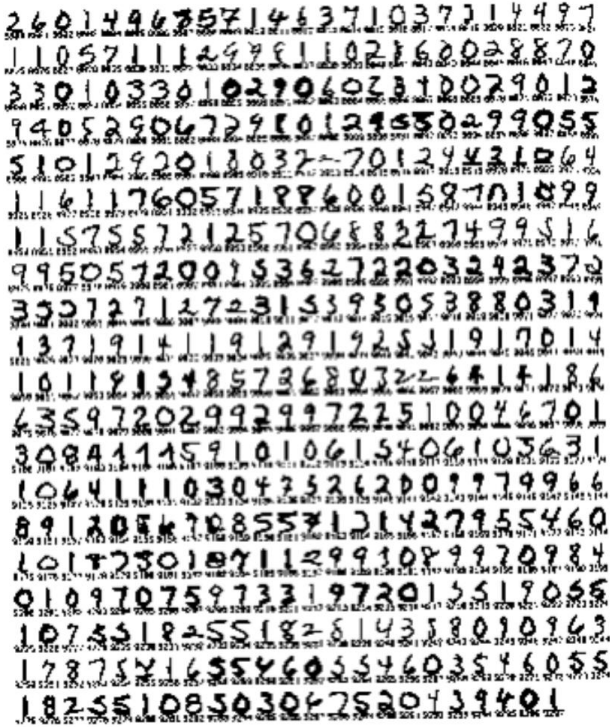
- Human performance: 2.5% error

Various learning algorithms (pre-deep learning

- 16.2%: Decision tree (C4.5)
- 5.9%: 2-layer neural network
- 5.1%: LeNet 1 - 5-layer neural network

Various SVM results

- 4.0%: Polynomial kernel (274 support vectors)
- 4.1%: Gaussian kernel



Handwritten Digit Clasification

Very little overfitting due to max-margin

degree of polynomial	dimensionality of feature space	support vectors	raw error
1	256	282	8.9
2	≈ 33000	227	4.7
3	$\approx 1 \times 10^6$	274	4.0
4	$\approx 1 \times 10^9$	321	4.2
5	$\approx 1 \times 10^{12}$	374	4.3
6	$\approx 1 \times 10^{14}$	377	4.5
7	$\approx 1 \times 10^{16}$	422	4.5

Recent results

- With more training data, better modeling of invariances, etc.
- Error down to about 0.5% with SVMs and 0.4% with neural networks

Takeaway messages

What have we learned today?

- Kernels estimate the similarity between samples
- They represent an inner product in a feature space
  - Allows to use potentially infinite dimensional
  - That's ok due to the kernel trick and regularization
- Many standard ML algorithms can be "kernelized"
  - I.e. rewritten in terms of inner products
  - **Regression:** Kernel Ridge regression, Gaussian Processes (to be covered), Support Vector Regression (not covered)
  - **Classification:** SVMs, Kernel Logistic Regression (not covered)

✓ Very flexible representation that adapts to the complexity of the data

✓ Works well with small data sets

× Hard to scale to more complex (high-D) problems or large datasets